

# Compton Scattering

## COMPTON SCATTERING

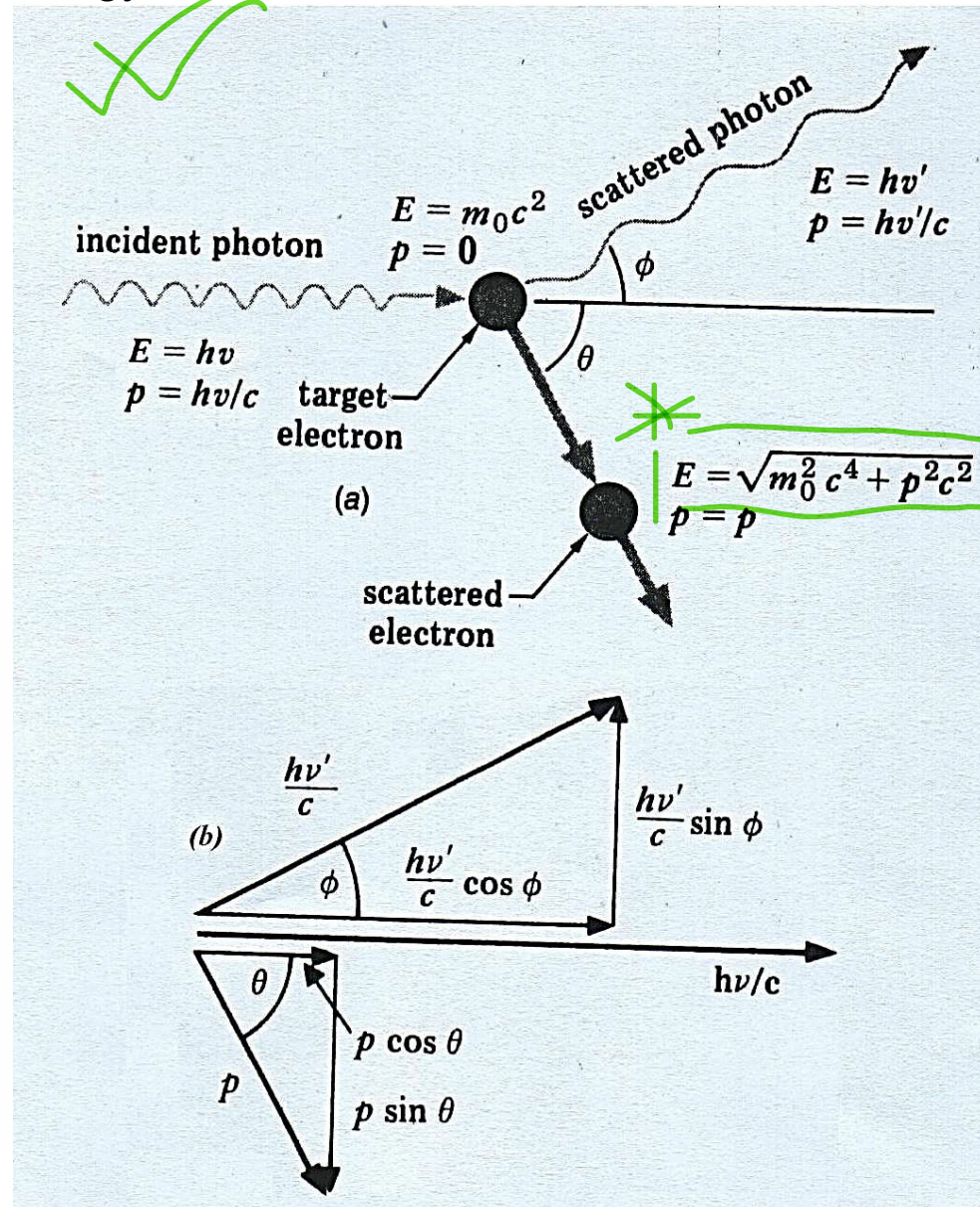
When a beam of intermediate-energy photon/x-rays of well defined wavelength  $\lambda$  is colliding with a target, then the loosely bound electrons are released from the outer shell of the atom or molecule of the target and some of the radiation/photon/x-rays is scattered in some angle  $\phi$  with a component of well defined wavelength  $\lambda'$ , which is longer than  $\lambda$ . The change in wavelength depends on the angle  $\theta$  through which the radiation is scattered. This phenomenon was first observed by A.H. Compton and is known as Compton effect.

The shift of wavelength is then  $\lambda' - \lambda = \frac{h}{m_0 c} [1 - \cos \phi]$

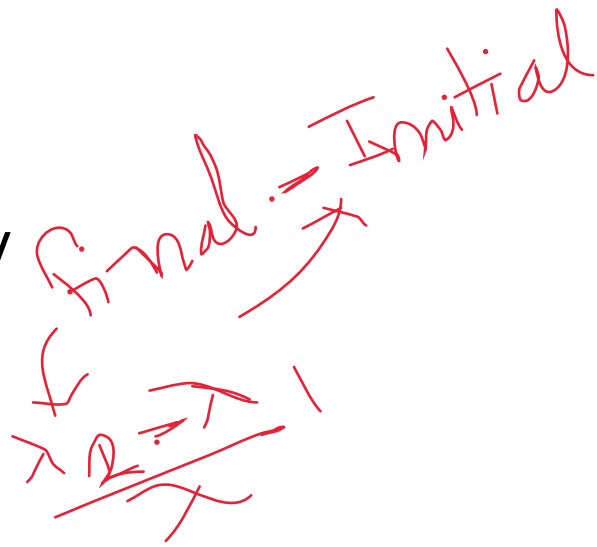
where,  $m_0$  is the electron rest mass. The quantity  $\frac{h}{m_0 c}$  has the unit of length and its value is:

$$\frac{h}{m_0 c} = \frac{6.626 \times 10^{-34} \text{ J.s}}{[9.109 \times 10^{-31} \text{ kg}][2.998 \times 10^8 \text{ m/s}]} = 2.426 \times 10^{-12} \text{ m}$$

The Compton scattering is simply the elastic scattering of a photon by an electron, in which both energy and momentum is conserved.



# Summary



• Fractional wavelength shift =

$$\frac{\Delta\lambda}{\lambda}$$

• Compton shift =

$$\lambda' - \lambda = \frac{h}{m_0 c} (1 - \cos \phi)$$

• Compton wavelength =

$$\lambda_c = \frac{h}{m_0 c}$$

Handwritten note:  $\rightarrow 2.426 \times 10^{-12} \text{ m}$

• Fractional energy loss =

$$\frac{E - E'}{E} = \frac{h\nu - h\nu'}{h\nu} = \frac{c/\lambda - c/\lambda'}{c/\lambda} = \frac{\lambda' - \lambda}{\lambda'} = \frac{\Delta\lambda}{\lambda + \Delta\lambda}$$

**Example 1:** Compare Compton scattering for x-rays ( $\lambda \approx 20$  pm) and visible light ( $\lambda \approx 500$  nm) at a particular angle of scattering. Which has greater [a] Compton shift, [b] fractional wavelength shift, [c] fractional photon energy change and [d] energy imparted to the electron.

[a] Compton shift,  $\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0 c} [1 - \cos \phi]$  is independent of wavelength and depends only on angle  $\phi$ . So both these waves have same shift in wavelength

$$\text{For } \phi = 90^\circ, \Delta\lambda = \frac{h}{m_0 c} [1 - \cos 90^\circ] = \frac{6.626 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^8} = 2.427 \times 10^{-12} \text{ m}$$

$$\text{[b] Fractional wavelength shift} = \frac{\Delta\lambda}{\lambda}$$

$$\text{For x-rays with } \lambda = 20 \text{ pm} \quad \frac{\Delta\lambda}{\lambda} = \frac{2.427 \times 10^{-12} \text{ m}}{20 \times 10^{-12} \text{ m}} = 0.12135$$

$$\text{For visible light with } \lambda = 500 \text{ nm} \quad \frac{\Delta\lambda}{\lambda} = \frac{2.427 \times 10^{-12} \text{ m}}{500 \times 10^{-9} \text{ m}} = 4.854 \times 10^{-6}$$



[c] Fractional energy loss =  $\frac{E - E'}{E} = \frac{\Delta\lambda}{\lambda + \Delta\lambda}$

For x-rays with  $\lambda = 20 \text{ pm}$   $\frac{E - E'}{E} = \frac{2.427 \times 10^{-12} \text{ m}}{20 \times 10^{-12} \text{ m} + 2.427 \times 10^{-12} \text{ m}} = 0.108$

For x-rays with  $\lambda = 500 \text{ nm}$   $\frac{E - E'}{E} = \frac{2.427 \times 10^{-12} \text{ m}}{500 \times 10^{-9} \text{ m} + 2.427 \times 10^{-12} \text{ m}} = 4.85 \times 10^{-6}$

[d] Energy imparted to electron  $E - E' = \frac{\Delta\lambda}{\lambda + \Delta\lambda} E$

For x-rays with  $\lambda = 20 \text{ pm}$   $E = \frac{hc}{\lambda} = \frac{1240 \text{ eV nm}}{20 \times 10^{-3} \text{ nm}} = 62000 \text{ eV} = 62 \text{ keV}$

Hence energy imparted to electron  $E - E' = 0.108 \times 62 \text{ keV} = 6.696 \text{ keV}$

For x-rays with  $\lambda = 500 \text{ nm}$   $E = \frac{hc}{\lambda} = \frac{1240 \text{ eV nm}}{500 \text{ nm}} = 2.48 \text{ eV}$

Hence energy imparted to electron  $E - E' = 4.85 \times 10^{-6} \times 2.48 \text{ eV} = 1.95 \times 10^{-6} \text{ eV}$

## DE BROGLIE WAVES/De Broglie's hypothesis:

We know that Photoelectric effect Compton effect could be explained if electromagnetic wave was supposed to consist of packets of energy called quanta or photon which behaved like a particle of rest mass 0 and moved through space with momentum  $p = mc$ , i.e., the photon moves with the velocity of light.

On the other hand, light exhibits the phenomena like interference, diffraction and polarization, which can only be explained by the wave nature of light. Thus, electromagnetic wave was regarded as exhibiting the behaviors of wave-particle duality. The physicist made use of either the wave or particle nature of light to explain the experimental result as suitable. This fact led Louis de Broglie in 1924 to make a fascinating suggestion that if light under certain circumstances behaves as waves and other circumstances behaves as particles, then particle of matter also should show similar behavior as that exhibited by the light.

We know that the energy of a photon of radiation having frequency  $\nu$  is given by

$$E = h\nu \quad \dots\dots \dots [1]$$

If the photon is considered to be a particle of mass  $m$  then its energy  $E = mc^2 = mc \cdot c = p \cdot c$ , where  $p$  is the momentum of photon.

Thus, we can write  $h\nu = pc \rightarrow \frac{c}{\nu} = \frac{h}{p} \rightarrow \lambda = \frac{h}{p} \dots\dots [2]$

de Broglie assumed that equation (2) should be equally applicable to the electromagnetic waves and the material particles. He considered that if a particle of mass  $m$  would move with a velocity of  $v$ , the wavelength of the wave associated with the particle should be given by

$$\lambda = \frac{h}{p} = \frac{h}{mv} \quad \dots \dots [3]$$

This wavelength of a particle is called de Broglie wavelength.

de Broglie's suggestion about the nature of particle was verified experimentally by Davisson and Garmer in 1927 and Thomson in 1928 and they showed that the electrons are diffracted when a beam of electrons fall on metal surface. Thus the truth of de Broglie's suggestion was established and he got the noble prize in 1929.

### Example 1:

Compute the de Broglie wavelength of the following:

[1] A 1000-kg automobile traveling at 25 m/s.

[2] A 10 gm bullet traveling at 500 m/s.

[3] A smoke particle of mass  $10^{-9}$  gm moving at 1cm/s.

[4] An electron with a K. E of 1keV.

[5] An electron with a K.E of 100 MeV.

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**Solution:**

$$1. \lambda = \frac{h}{p} = \frac{h}{mv} = \frac{6.6 \times 10^{-34} \text{ J.s}}{[1000 \text{ kg}][25 \text{ m/s}]} = 2.6504 \times 10^{-38} \text{ m}$$

$$2. \lambda = \frac{h}{p} = \frac{h}{mv} = \frac{6.6 \times 10^{-34} \text{ J.s}}{[10^{-2} \text{ kg}][500 \text{ m/s}]} = 1.3 \times 10^{-34} \text{ m}$$

$$3. \lambda = \frac{h}{p} = \frac{h}{mv} = \frac{6.6 \times 10^{-34} \text{ J.s}}{[10^{-12} \text{ kg}][10^{-2} \text{ m/s}]} = 6.6 \times 10^{-20} \text{ m}$$

$$4. \quad \lambda = \frac{h}{p} = \frac{hc}{pc} = \frac{hc}{\sqrt{E^2 - m_0^2 c^4}} = \frac{hc}{\sqrt{(KE + m_0 c^2)^2 - m_0^2 c^4}} = \frac{1240 \text{ eV} \cdot \text{nm}}{31984 \text{ eV}} = 0.0387 \text{ nm}$$

$$5. \quad \lambda = \frac{hc}{pc} = \frac{hc}{\sqrt{(KE + m_0 c^2)^2 - m_0^2 c^4}} = \frac{1240 \text{ eV} \cdot \text{nm}}{1.00498 \times 10^8 \text{ eV}} = 0.12338 \times 10^{-15} \text{ m} = 0.12338 \text{ fm}$$

Note that the wavelengths computed in parts [1] to [3] are far too small to be observed in the laboratory. Only in the last two cases, in which the wavelength is of the same order as atomic or nuclear sizes, we have the chance of observing the wavelength.

Because of the smallness of  $h$ , only for particles of atomic or nuclear size will the wave behavior be observable.

$$E^2 = p^2 c^2 + m_0^2 c^4$$

**Example 2:**

An electron has a de Broglie wavelength of  $2.00 \text{ pm} = 2.00 \times 10^{-12} \text{ m}$ . Find its K.E, phase velocity and group velocity of its de Broglie waves.

**Solution:**

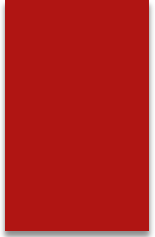
$$pc = \frac{hc}{\lambda} = \frac{[4.136 \times 10^{-15} \text{ eV.s}][3.00 \times 10^8 \text{ m/s}]}{2.00 \times 10^{-12} \text{ m}} = 6.20 \times 10^5 \text{ eV} = 620 \text{ keV}$$

Rest energy of the electron  $= E_0 = 511 \text{ keV}$

Therefore,

$$\begin{aligned} \text{K.E} = E - E_0 &= \sqrt{E_0^2 + (pc)^2} - E_0 = \sqrt{(511 \text{ keV})^2 + (600 \text{ keV})^2} - 511 \text{ keV} \\ &= 803 \text{ keV} - 511 \text{ keV} = 292 \text{ keV} \end{aligned}$$

The electron velocity can be found from 
$$E = \frac{E_0}{\sqrt{1 - v^2/c^2}}$$


$$v = c\sqrt{1 - E_0^2/E^2} = c\sqrt{1 - [511/803]^2} = 0.77c$$

Therefore,

$$v_p = \frac{c^2}{v} = \frac{c^2}{0.77c} = 1.30c$$

$$v_g = v$$

### Example 3:

Find the de Broglie wavelength of (a) 46-g golf ball with a velocity of 30 m/s, and (b) an electron with a velocity of  $10^7$  m/s.

#### Solution:

Since  $v \ll c$ , we can let  $m = m_0$ . Hence

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34} \text{ J.s}}{(0.046 \text{ kg})(30 \text{ m/s})} = 4.8 \times 10^{-34} \text{ m}$$

The wavelength of the golf ball is so small compared with its dimensions that we would not expect to find any wave aspects in its behaviour.

(b) Again  $v \ll c$ , so  $m = m_0 = 9.1 \times 10^{-31}$  kg for the electron.

$$\lambda = \frac{h}{m_0 v} = \frac{6.63 \times 10^{-34} \text{ J.s}}{(9.1 \times 10^{-31} \text{ kg})(10^7 \text{ m/s})} = 7.3 \times 10^{-11} \text{ m}$$